day_2_Debarshi

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2026-07-24

Define the directory path containing our CEL files

```
celpath <- "/home/ibab/bio-env"
```

Load packages for microarray processing

```
library(affy)
library(limma)
library(tidyverse)
```

Read raw microarray data from the specified directory

```
yeast_microchip <- ReadAffy(celfile.path = celpath, compress = TRUE)
```

Extract gene expression matrix into a standard data frame

```
df_yeast <- data.frame(exprs(yeast_microchip))
```

Preview the top rows of the expression matrix

```
head(df_yeast)
```

	EWE11465_0001_WT1.CEL <dbl>	EWE11465_0002_WT2.CEL <dbl>	EWE11465_0003_WT3.CEL <dbl>	EWE11465
1	75	95	83	
2	4530	4222	4416	
3	74	91	78	
4	4918	4362	4714	

	EWE11465_0001_WT1.CEL <dbl>	EWE11465_0002_WT2.CEL <dbl>	EWE11465_0003_WT3.CEL <dbl>	EWE11465
5	111	79	80	
6	66	87	77	

6 rows | 1-5 of 7 columns

Extract Perfect Match (PM) probe signals

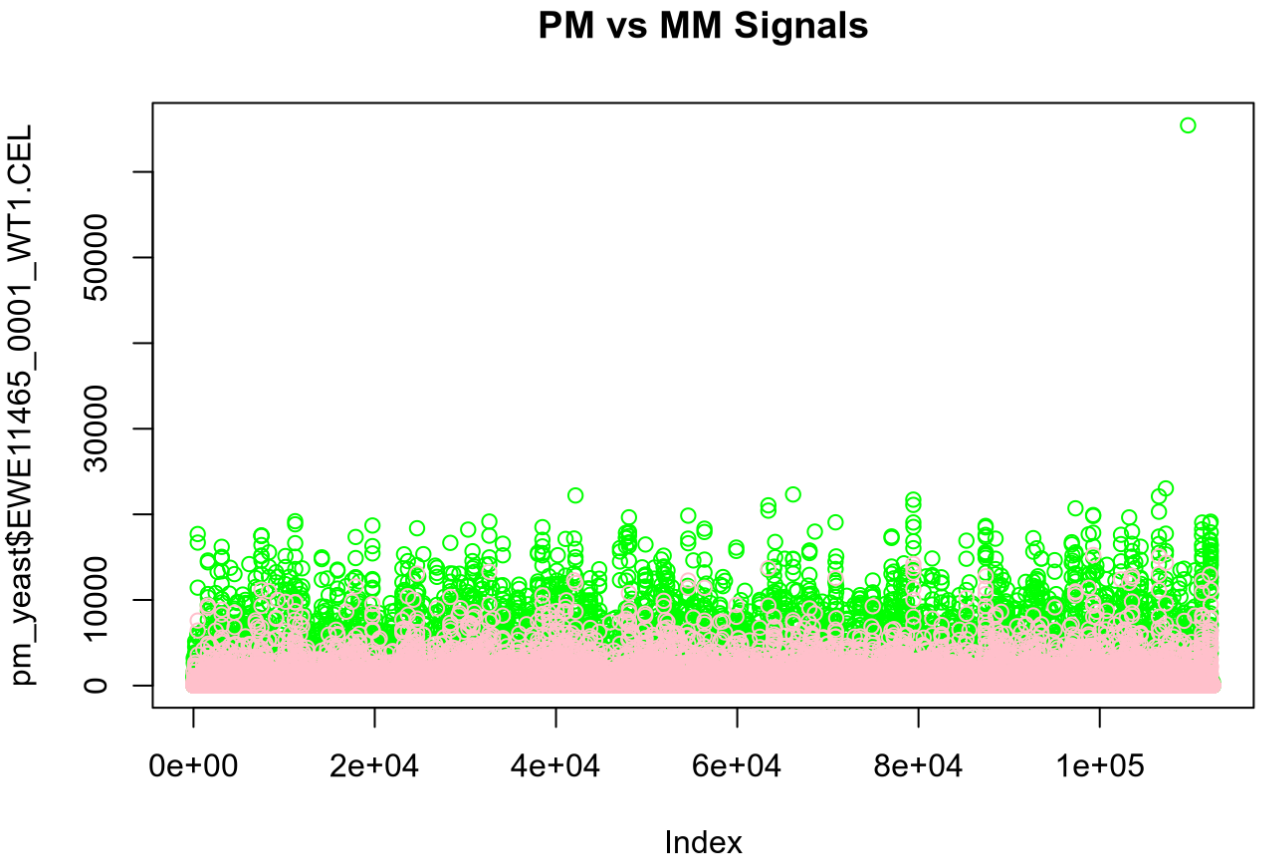
```
pm_yeast <- data.frame(affy::pm(yeast_microchip))
```

Extract Mismatch (MM) probe signals

```
mm_yeast <- data.frame(affy::mm(yeast_microchip))
```

Scatter plot of PM intensities for the first Wild Type sample

```
plot(pm_yeast$EWE11465_0001_WT1.CEL, col = "green", main = "PM vs MM Signals")
# Overlay MM intensities on the same plot
points(mm_yeast$EWE11465_0001_WT1.CEL, col = "pink")
```



Performed log-transformation on PM and MM intensity data

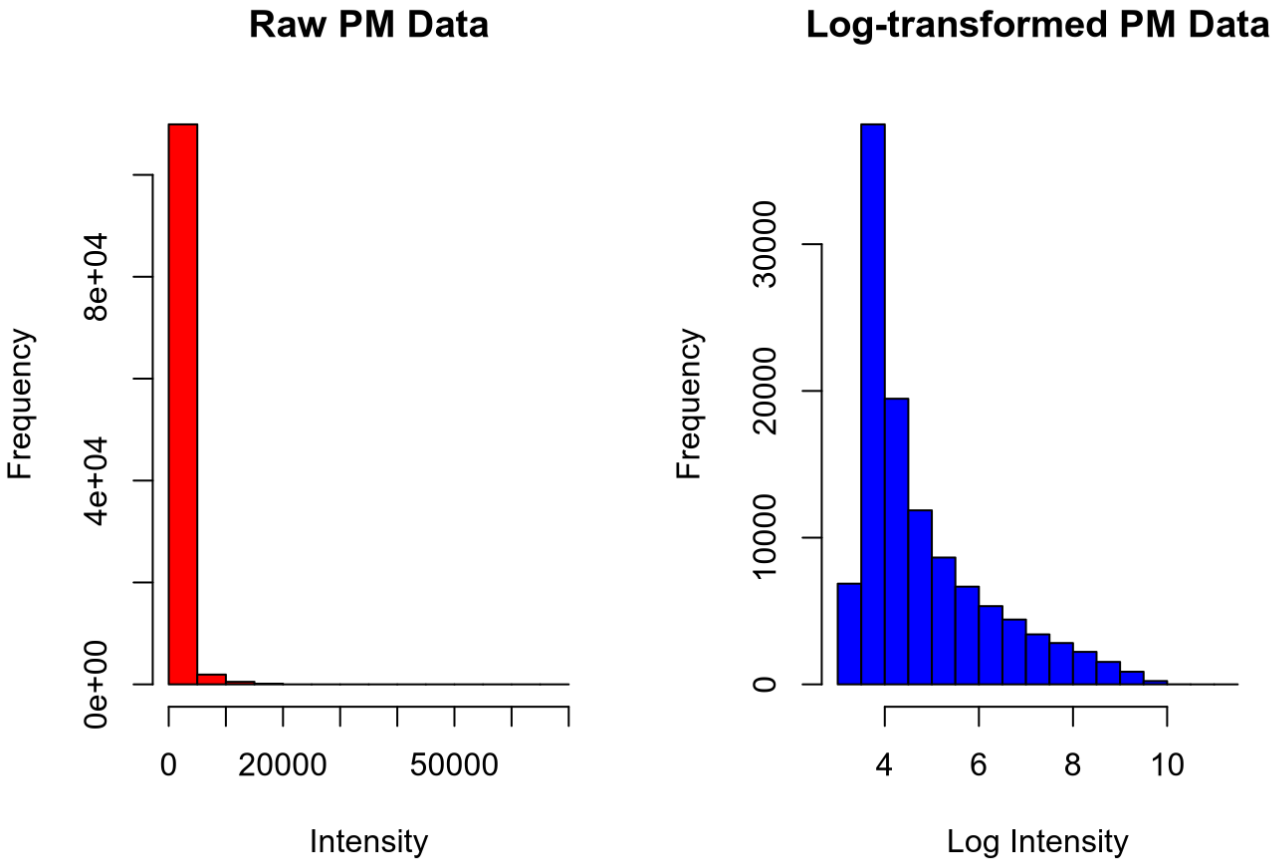
```
pmlog <- log(pm_yeast)
mmlog <- log(mm_yeast)
```

Set up layout for two side-by-side plots

```
par(mfrow = c(1, 2))

# Histogram of raw PM values
hist(pm_yeast$EWE11465_0001_WT1.CEL, main = "Raw PM Data", xlab = "Intensity", col
= "red")

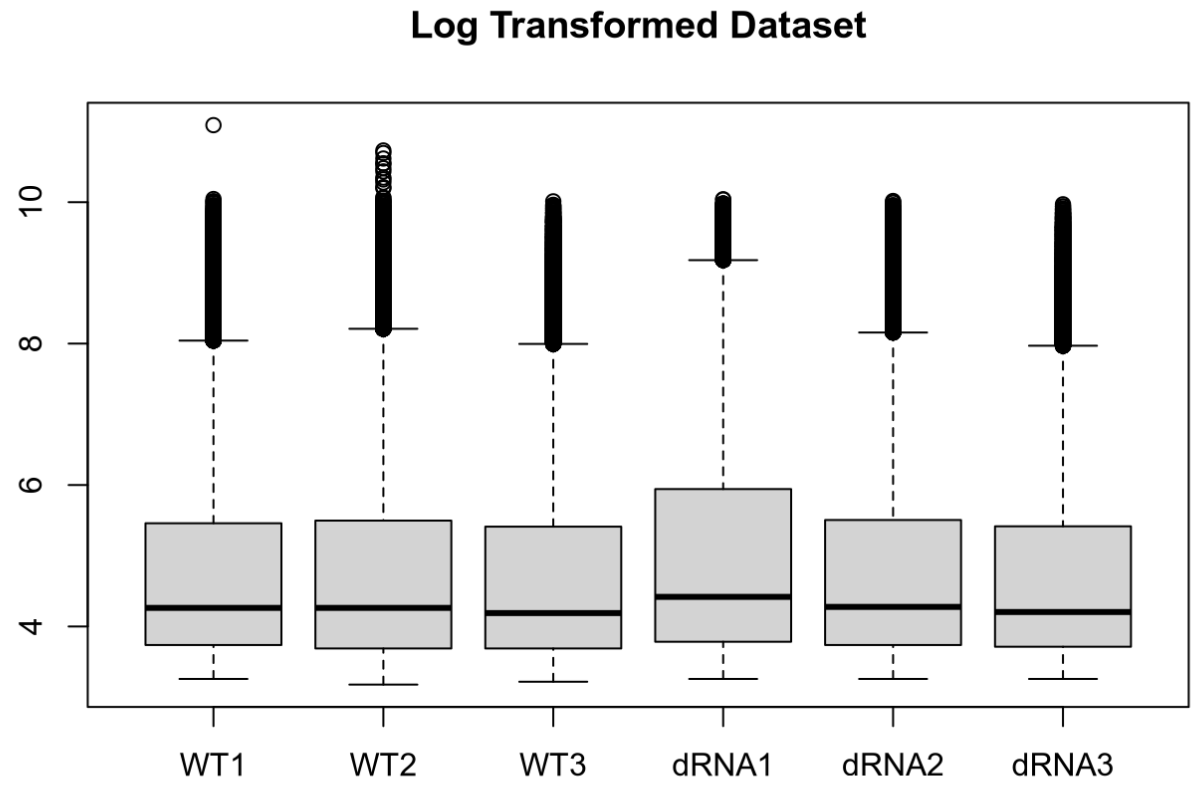
# Histogram of log-transformed PM values
hist(pmlog$EWE11465_0001_WT1.CEL, main = "Log-transformed PM Data", xlab = "Log In
tensity", col = "blue")
```



Extract individual sample log-PM vectors

```
wt1_log  <- pmlog$EWE11465_0001_WT1.CEL
wt2_log  <- pmlog$EWE11465_0002_WT2.CEL
wt3_log  <- pmlog$EWE11465_0003_WT3.CEL
drna1_log <- pmlog$EWE11465_0004_drna1.CEL
drna2_log <- pmlog$EWE11465_0005_drna2.CEL
drna3_log <- pmlog$EWE11465_0006_drna3.CEL

# Plot boxplots across samples
boxplot(wt1_log, wt2_log, wt3_log, drna1_log, drna2_log, drna3_log,
        names = c("WT1", "WT2", "WT3", "dRNA1", "dRNA2", "dRNA3"),main = "Log Tran
sformed Dataset")
```



Applied RMA

```
yeast_microchip_rma <- rma(yeast_microchip)
```

```
## Background correcting
## Normalizing
## Calculating Expression
```

Extraced gene expression matrix into a standard data frame

```
df_rma_yeast <- data.frame(exprs(yeast_microchip_rma))

#the top rows of the expression matrix
head(df_rma_yeast)
```

	EWE11465_0001_WT1.CEL <dbl>	EWE11465_0002_WT2.CEL <dbl>	EWE11465_0003_WT3.CE <db
1759068_at	3.413791	3.015579	3.43726
1759069_at	9.020832	9.126874	9.04889
1759070_s_at	7.771956	7.623827	7.43295
1759071_s_at	6.148551	5.695259	5.83669
1759072_s_at	3.670370	3.580104	3.66205
1759073_at	2.925485	3.253104	2.65179

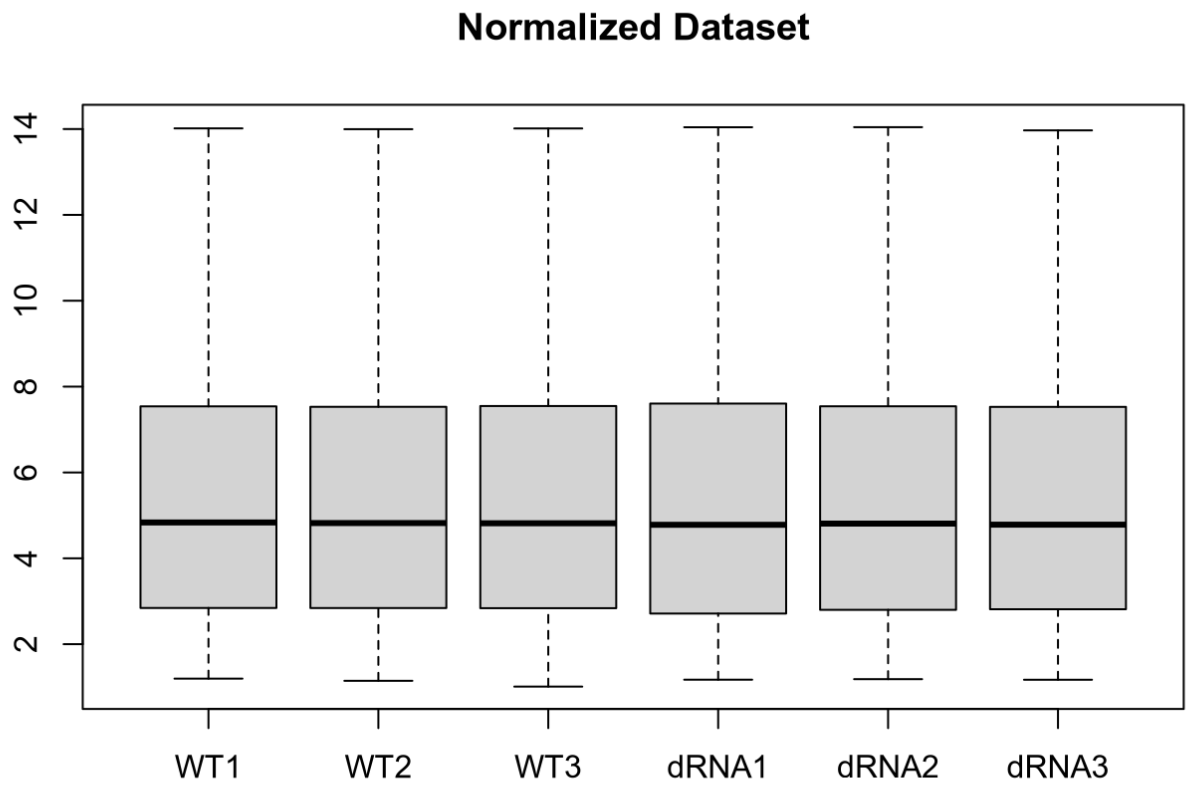
6 rows | 1-4 of 7 columns

Stored individual sample vectors from the RMA-normalized data frame

```
wt1_norm  <- df_rma_yeast$EWE11465_0001_WT1.CEL
wt2_norm  <- df_rma_yeast$EWE11465_0002_WT2.CEL
wt3_norm  <- df_rma_yeast$EWE11465_0003_WT3.CEL
drna1_norm <- df_rma_yeast$EWE11465_0004_drna1.CEL
drna2_norm <- df_rma_yeast$EWE11465_0005_drna2.CEL
drna3_norm <- df_rma_yeast$EWE11465_0006_drna3.CEL
```

Boxplot of RMA-normalized expression values

```
boxplot(wt1_norm, wt2_norm, wt3_norm, drna1_norm, drna2_norm, drna3_norm,
        names = c("WT1", "WT2", "WT3", "dRNA1", "dRNA2", "dRNA3"),main = "Normaliz
ed Dataset")
```



Prepared raw vectors for comparison

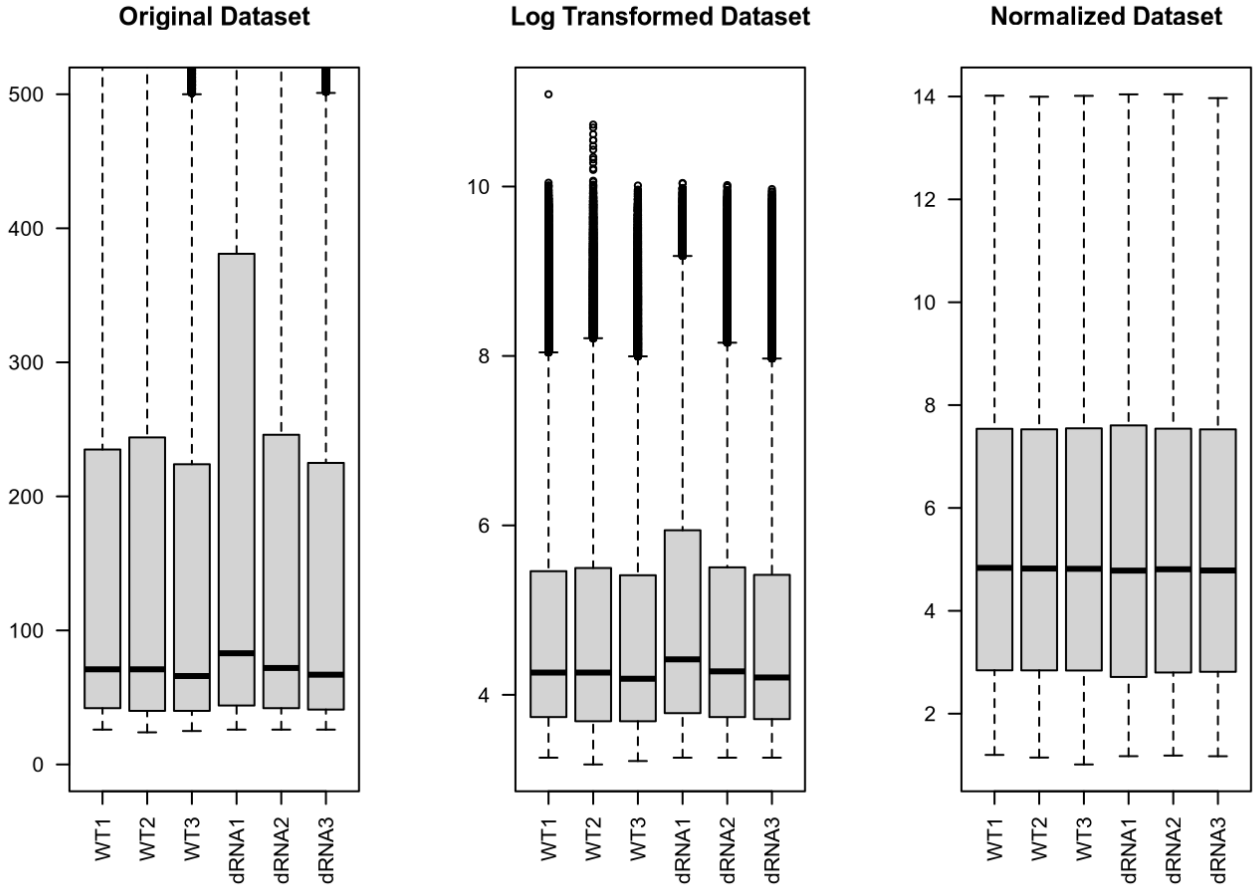
```
wt1_og  <- pm_yeast$EWE11465_0001_WT1.CEL
wt2_og  <- pm_yeast$EWE11465_0002_WT2.CEL
wt3_og  <- pm_yeast$EWE11465_0003_WT3.CEL
drna1_og <- pm_yeast$EWE11465_0004_drna1.CEL
drna2_og <- pm_yeast$EWE11465_0005_drna2.CEL
drna3_og <- pm_yeast$EWE11465_0006_drna3.CEL

# Display three side-by-side boxplots
par(mfrow = c(1, 3))

# 1. Original raw dataset
boxplot(wt1_og, wt2_og, wt3_og, drna1_og, drna2_og, drna3_og,
        names = c("WT1", "WT2", "WT3", "dRNA1", "dRNA2", "dRNA3"),
        ylim = c(0, 500), main = "Original Dataset", las = 2)

# 2. Log-transformed dataset
boxplot(wt1_log, wt2_log, wt3_log, drna1_log, drna2_log, drna3_log,
        names = c("WT1", "WT2", "WT3", "dRNA1", "dRNA2", "dRNA3"),main = "Log Tran
sformed Dataset", las = 2)

# 3. RMA-normalized dataset
boxplot(wt1_norm, wt2_norm, wt3_norm, drna1_norm, drna2_norm, drna3_norm,
        names = c("WT1", "WT2", "WT3", "dRNA1", "dRNA2", "dRNA3"),main = "Normaliz
ed Dataset", las = 2)
```



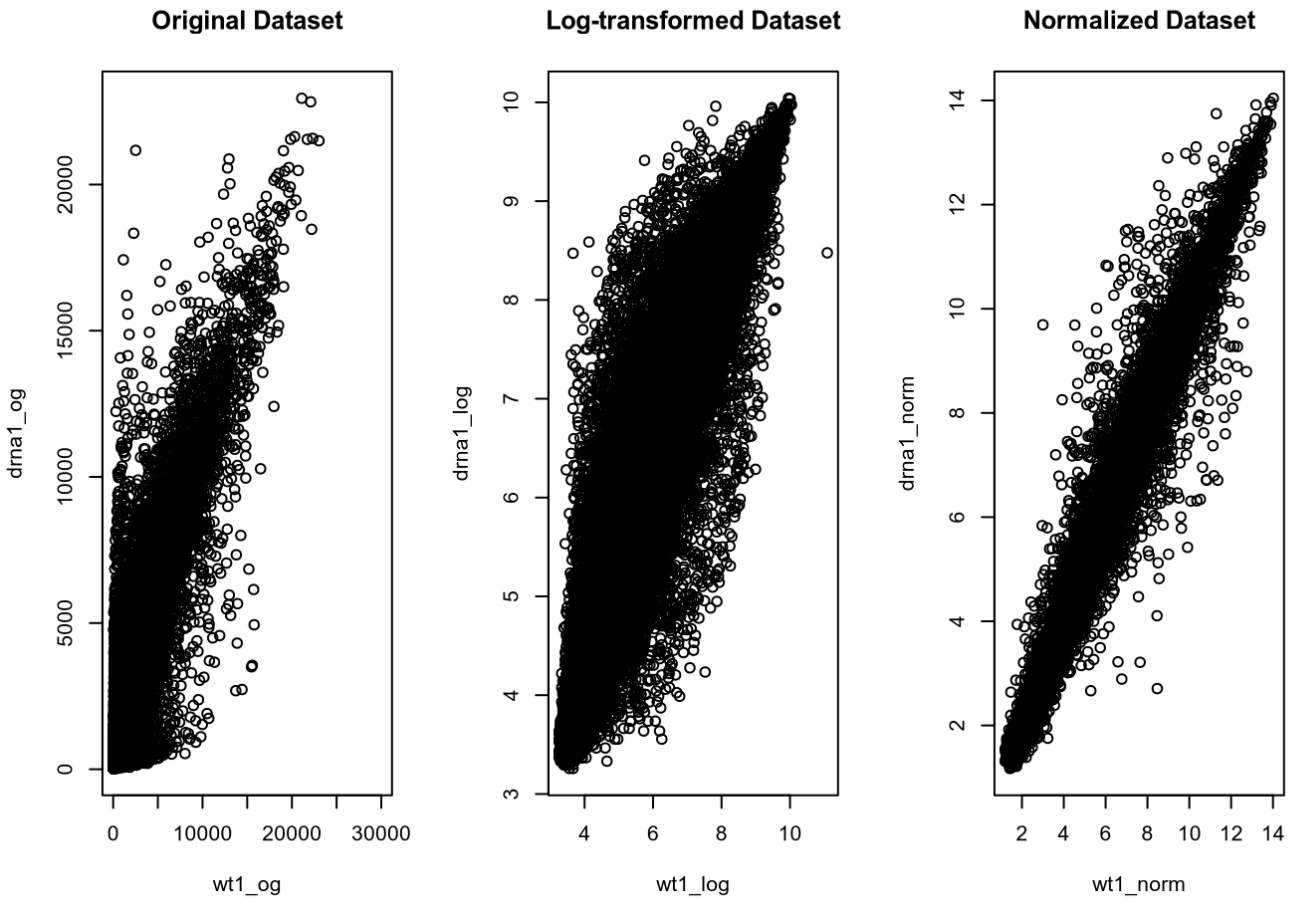
#compared scatter plots

```
par(mfrow = c(1, 3))

# Raw scatter plot
plot(wt1_og, drna1_og, main = "Original Dataset", xlim = c(0, 30000))

# Log-transformed scatter plot
plot(wt1_log, drna1_log, main = "Log-transformed Dataset")

# Normalized scatter plot
plot(wt1_norm, drna1_norm, main = "Normalized Dataset")
```



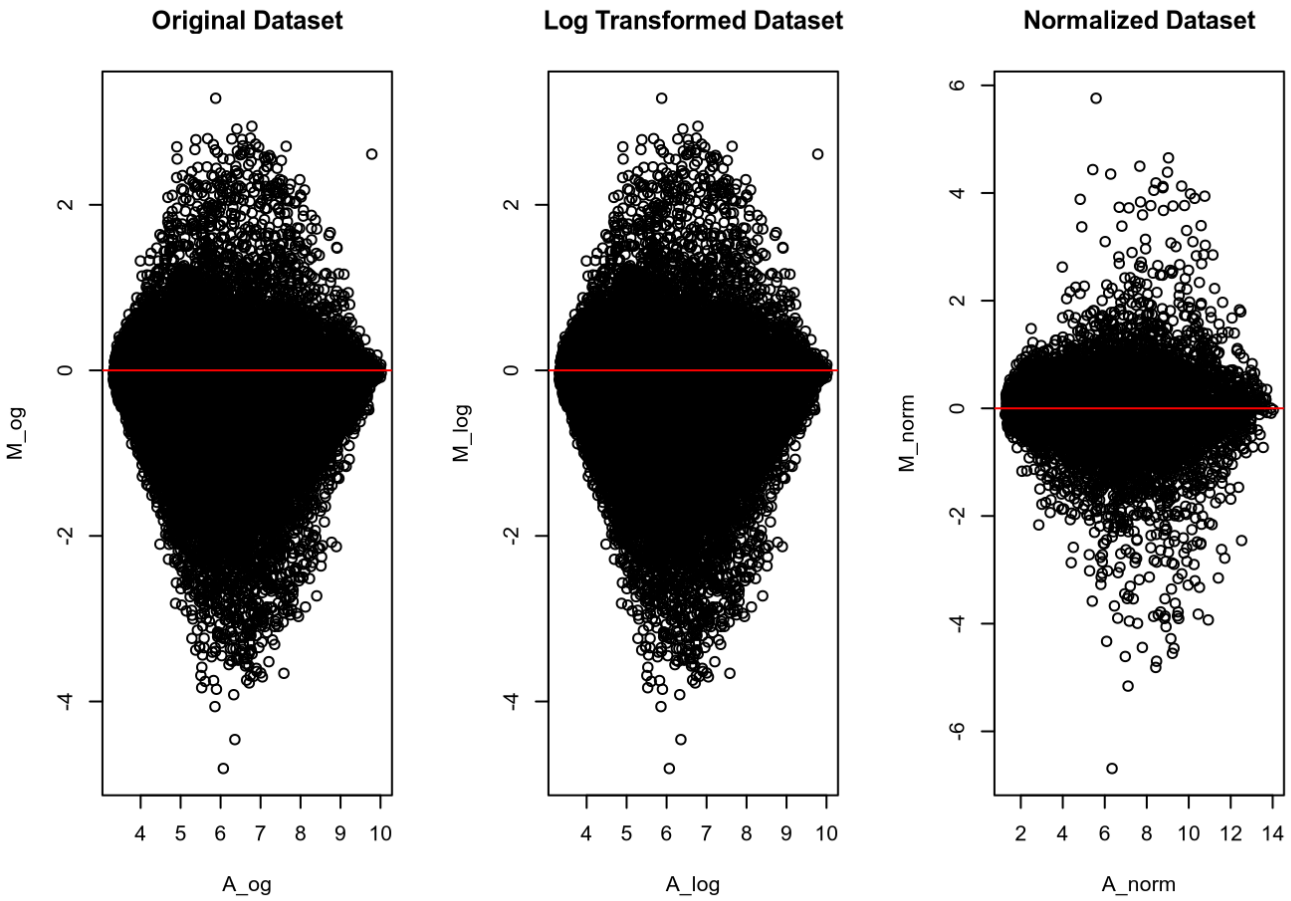
Generated MA plots

```
par(mfrow = c(1, 3))

# MA Calculation: Raw Data
A_og <- (log(wt1_og) + log(drna1_og)) / 2
M_og <- log(wt1_og / drna1_og)
plot(A_og, M_og, main = "Original Dataset")
abline(h = 0, col = "red")

# MA Calculation: Log-Transformed Data
A_log <- (wt1_log + drna1_log) / 2
M_log <- wt1_log - drna1_log
plot(A_log, M_log, main = "Log Transformed Dataset")
abline(h = 0, col = "red")

# MA Calculation: Normalized Data
A_norm <- (wt1_norm + drna1_norm) / 2
M_norm <- wt1_norm - drna1_norm
plot(A_norm, M_norm, main = "Normalized Dataset")
abline(h = 0, col = "red")
```


Original Dataset
M_og
A_og
Log Transformed Dataset
M_log
A_log
Normalized Dataset
M_norm
A_norm